

Sets

Sets

Definition 1: A *set* is an unordered collection of objects.

Definition 2:

The objects in a set are called the *elements*, or *members*, of the set.

We write $a \in A$ to denote that a is an element of the set A .

The notation $a \notin A$ denotes that a is not a member of the set A .

A set is described by listing all the members of the set.

Example: The set V of all vowels in the English alphabet can be written as $V = \{a, e, i, o, u\}$.

Example: The set O of odd positive integers less than 10 can be expressed by $O = \{1, 3, 5, 7, 9\}$.

Example: $\{a, 2, \text{Fred}, \text{New Jersey}\}$ is the set containing the four elements a , 2, Fred, and New Jersey.

Example: The set of positive integers less than 100 can be denoted by $\{1, 2, 3, \dots, 99\}$.

Another way of describing a set is to use **set builder notation**. We characterize all those elements in the set by stating the property or properties they must have to be members.

The set O of all odd positive integers less than 10 can be written as

$$O = \{x \mid x \text{ is an odd positive integer less than } 10\}$$

These sets play an important role in discrete mathematics:

$\mathbf{N} = \{0, 1, 2, 3, \dots\}$, the set of **natural numbers**

$\mathbf{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$, the set of **integers**

$\mathbf{Z}^+ = \{1, 2, 3, \dots\}$, the set of **positive integers**

$\mathbf{Q} = \{p/q \mid p \in \mathbf{Z}, q \in \mathbf{Z}, q \neq 0\}$, the set of **rational numbers**

$\mathbf{R}$, the set of **real numbers**

Definition 3: Two sets are *equal* if and only if they have the same elements. That is, if A and B are sets, then A and B are equal if and only if $\forall x(x \in A \leftrightarrow x \in B)$. We write $A = B$ if A and B are equal sets.

Example: The sets $A = \{1, 3, 5\}$ and $B = \{3, 5, 1\}$ are equal.

Example: The sets $A = \{1, 3, 5\}$ and $B = \{3, 5, 1, 3, 5\}$ are equal.

Sets can be represented graphically using Venn diagrams.

In Venn diagrams the **universal set** U , which contains all the objects under consideration, is represented by a rectangle.

Inside this rectangle, circles are used to represent sets.

Sometimes points are used to represent the particular elements of the set.

Example: Draw a Venn diagram that represents V , the set of vowels in the English alphabet.

Solution:

Universal set U is the set of the 26 letters of the English alphabet.

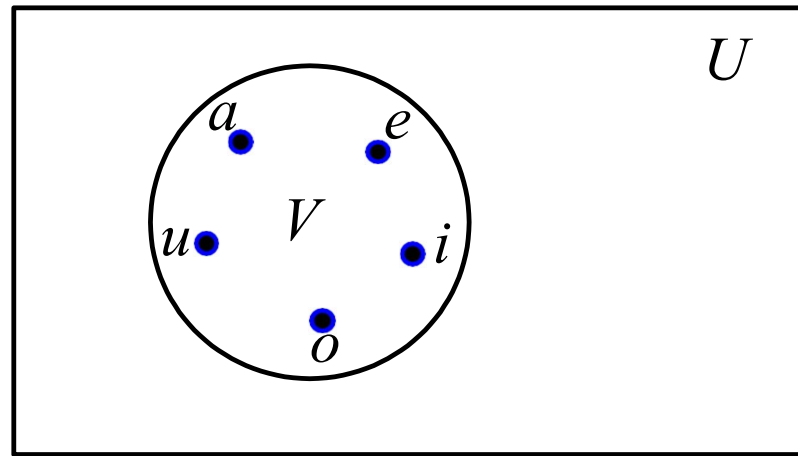


Figure 1 Venn Diagram for the Set of Vowels.

There is a special set that has no elements. This set is called the **empty set**, or **null set**, and is denoted by $\emptyset$. The empty set can also be denoted by $\{ \}$.

A set with one element is called a **singleton set**.

Definition 4: The set A is said to be a *subset* of B if and only if every element of A is also an element of B . We use the notation $A \subseteq B$ to indicate that A is a subset of the set B .

We see that $A \subseteq B$ if and only if $\forall x(x \in A \rightarrow x \in B)$ is true.

Example:

The set of all odd positive integers less than 10 is a subset of the set of all positive integers less than 10.

The set of rational numbers is a subset of the set of real numbers.

Theorem 1: For any set S , (i) $\emptyset \subseteq S$ and (ii) $S \subseteq S$.

If a set has n distinct elements, then it has 2^n subsets.

Example:

$$A = \{a, b\}$$

Subsets are $\emptyset$, $\{a\}$, $\{b\}$, $\{a, b\}$

When we wish to emphasize that a set A is a subset of the set B but $A \neq B$, we write $A \subset B$ and say that A is a **proper subset** of B .

$A \subset B$ if and only if $\forall x(x \in A \rightarrow x \in B) \wedge \exists x(x \in B \wedge x \notin A)$ is true.

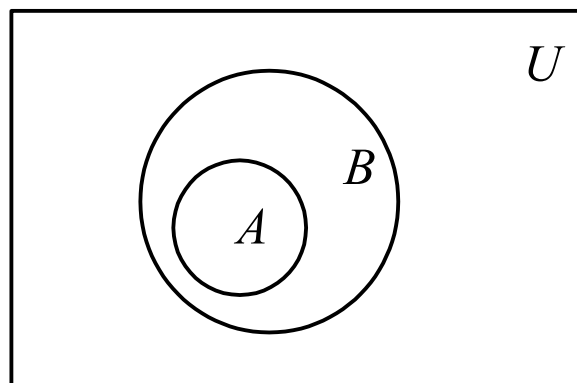


Figure: Venn Diagram Showing that A is a proper subset of B.

Sets may have other sets as member.

$$A = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$$

$$B = \{x \mid x \text{ is a subset of the set } \{a, b\}\}$$

These two sets are equal, that is, $A = B$.

Note that $\{a\} \in A$, but $a \notin A$.

Definition 5: Let S be a set. If there are exactly n distinct elements in S where n is a nonnegative integer, we say that S is a *finite set* and that n is the *cardinality* of S . The cardinality of S is denoted by $|S|$.

Example: Let A be the set of odd positive integers less than 10, that is, $A = \{1, 3, 5, 7, 9\}$. Then $|A| = 5$.

Example: Let S be the set of letters in the English alphabet. Then $|S| = 26$.

Example: Since the null set has no elements, $|\emptyset| = 0$.

Definition 6: A set is said to be *infinite* if it is not finite.

Example: The set of positive integers is infinite.

The Power Set

Definition 7: Given a set S , the *power set* of S is the set of all subsets of the set S . The power set of S is denoted by $P(S)$.

If a set has n distinct elements, then its power set has 2^n elements.

Example: What is the power set of the set $\{0, 1, 2\}$?

Solution:

$$\begin{aligned} P(\{0, 1, 2\}) \\ = \{\emptyset, \{0\}, \{1\}, \{2\}, \{0, 1\}, \{0, 2\}, \{1, 2\}, \{0, 1, 2\}\} \end{aligned}$$

Example: What is the power set of the empty set? What is the power set of the set $\{\emptyset\}$?

Solution:

$$\begin{aligned} P(\emptyset) &= \{\emptyset\}. \\ P(\{\emptyset\}) &= \{\emptyset, \{\emptyset\}\} \end{aligned}$$

Cartesian Products

Definition 8: The *ordered n -tuple* $(a_1, a_2, \dots, a_n)$ is the ordered collection that has a_1 as its first element, a_2 as its second element, ..., and a_n as its n th element.

We say that two ordered n -tuples are equal if and only if each corresponding pair of their elements is equal. In other words, $(a_1, a_2, \dots, a_n) = (b_1, b_2, \dots, b_n)$ if and only if $a_i = b_i$, for $i = 1, 2, \dots, n$.

2-tuples are called **ordered pairs**.

The ordered pairs (a, b) and (c, d) are equal if and only if $a = c$ and $b = d$.

(a, b) and (b, a) are not equal unless $a = b$.

Definition 9: Let A and B be sets. The Cartesian product of A and B , denoted by $A \times B$, is the set of all ordered pairs (a, b) where $a \in A$ and $b \in B$.

Hence, $A \times B = \{(a, b) \mid a \in A \wedge b \in B\}$.

Example: What is the Cartesian product of $A = \{1, 2\}$ and $B = \{a, b, c\}$?

Solution:

$$A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}$$

A subset R of the Cartesian product $A \times B$ is called a **relation** from the set A to the set B .

The elements of R are ordered pairs, where the first element belongs to A and the second to B .

For example,

$$R = \{(a, 0), (a, 1), (a, 3), (b, 1), (b, 2), (c, 0), (c, 3)\}$$

is a relation from the set $\{a, b, c\}$ to the set $\{0, 1, 2, 3\}$.

The Cartesian product $A \times B$ and $B \times A$ are not equal, unless $A = \emptyset$ or $B = \emptyset$ (so that $A \times B = \emptyset$) or unless $A = B$.

Example: Show that the Cartesian product $B \times A$ is not equal to the Cartesian product $A \times B$, where $A = \{1, 2\}$ and $B = \{a, b, c\}$.

Solution:

$$B \times A = \{(a, 1), (a, 2), (b, 1), (b, 2), (c, 1), (c, 2)\}$$

$$A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}$$

Therefore, $B \times A \neq A \times B$.

Example: What is the Cartesian product $A \times B \times C$, where $A = \{0, 1\}$, $B = \{1, 2\}$, and $C = \{0, 1, 2\}$?

Solution:

$$A \times B$$

$$= \{0, 1\} \times \{1, 2\}$$

$$= \{(0, 1), (0, 2), (1, 1), (1, 2)\}$$

$$(A \times B) \times C$$

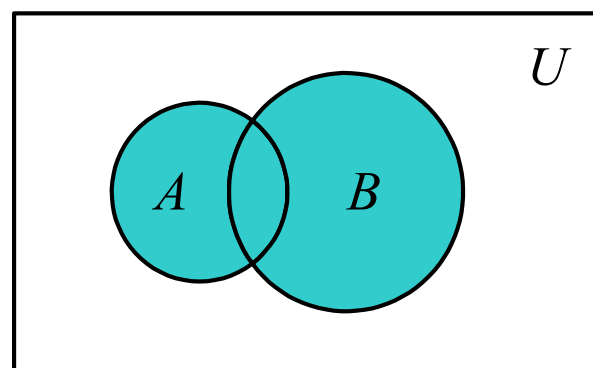
$$= \{(0, 1), (0, 2), (1, 1), (1, 2)\} \times \{0, 1, 2\}$$

$$= \{(0, 1, 0), (0, 1, 1), (0, 1, 2), \\ (0, 2, 0), (0, 2, 1), (0, 2, 2), \\ (1, 1, 0), (1, 1, 1), (1, 1, 2), \\ (1, 2, 0), (1, 2, 1), (1, 2, 2)\}$$

Set Operations

Definition 1: Let A and B be sets. The *union* of the sets A and B , denoted by $A \cup B$, is the set that contains those elements that are in A or in B , or in both.

$$A \cup B = \{x \mid x \in A \vee x \in B\}.$$



$A \cup B$ is shaded

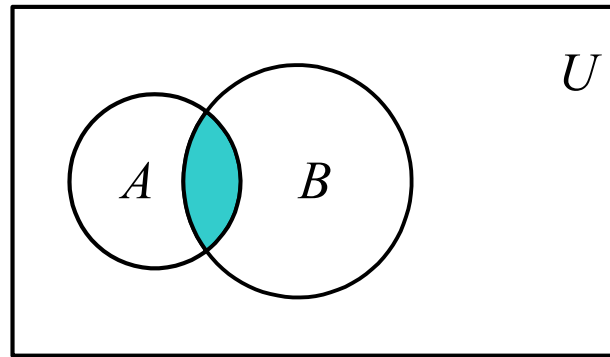
Figure: Venn Diagram Representing the Union of A and B .

Example: $A = \{1, 3, 5\}$ and $B = \{1, 2, 3\}$

$$A \cup B = \{1, 2, 3, 5\}$$

Definition 2: Let A and B be sets. The *intersection* of the sets A and B , denoted by $A \cap B$, is the set that contains those elements in both A and B .

$$A \cap B = \{x \mid x \in A \wedge x \in B\}.$$



$A \cap B$ is shaded

Figure: Venn Diagram Representing the Intersection of A and B .

Example: $A = \{1, 3, 5\}$ and $B = \{1, 2, 3\}$

$$A \cap B = \{1, 3\}$$

Definition 3: Two sets are called *disjoint* if their intersection is the empty set.

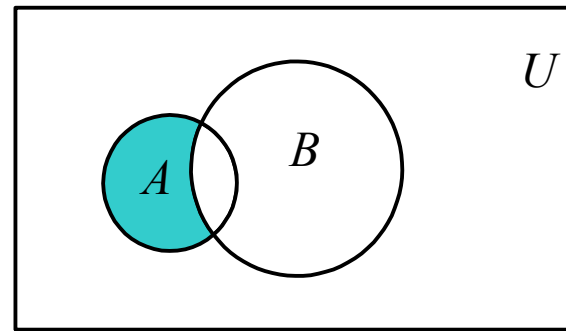
Example: Let $A = \{1, 3, 5, 7, 9\}$ and $B = \{2, 4, 6, 8, 10\}$. Since $A \cap B = \emptyset$, A and B are disjoint.

Definition 4: Let A and B be sets. The difference of A and B , denoted by $A-B$, is the set containing those elements that are in A but not in B .

$$A-B = \{x \mid x \in A \wedge x \notin B\}$$

The difference of A and B is also called the *complement of B with respect to A* .

$$A-B = A \cap \bar{B}$$



$A-B$ is shaded.

Figure: Venn Diagram for the Difference of A and B .

Example:

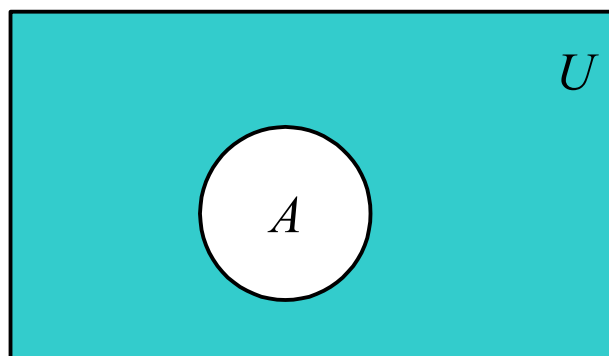
$$A = \{1, 3, 5\} \text{ and } B = \{1, 2, 3\}. A-B = \{5\}$$

$$A = \{1, 2, 3\} \text{ and } B = \{1, 3, 5\}. A-B = \{2\}$$

Definition 5: Let U be the universal set. The complement of the set A , denoted by A^c , is the complement of A with respect to U .

$$\bar{A} = \{x \mid x \notin A\}$$

In other words, $\bar{A} = U - A$.



$\bar{A}$ is shaded.

Figure: Venn Diagram for the Complement of the Set A .

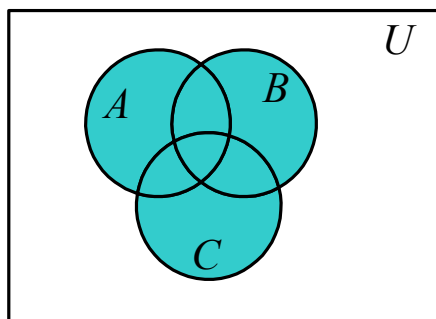
Example: Let $A = \{a, e, i, o, u\}$ (where the universal set is the set of letters of the English alphabet). Then $\bar{A} = \{b, c, d, f, g, h, j, k, l, m, n, p, q, r, s, t, v, w, x, y, z\}$

Example: Let A be the set of positive integers greater than 10 (with universal set the set of all positive integers). Then $\bar{A} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$.

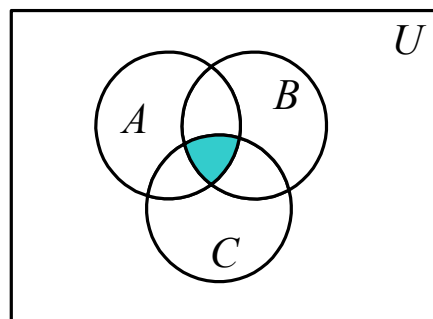
Generalized Unions and Intersections

$A \cup B \cup C$ contains those elements that are in at least one of the sets A , B , and C .

$A \cap B \cap C$ contains those elements that are in all of sets A , B , and C .



(a) $A \cup B \cup C$ is shaded.



(b) $A \cap B \cap C$ is shaded.

Figure: The Union and Intersection of A , B , and C .

Example:

Let $A = \{0, 2, 4, 6, 8\}$, $B = \{0, 1, 2, 3, 4\}$, and $C = \{0, 3, 6, 9\}$. What are $A \cup B \cup C$ and $A \cap B \cap C$?

Solution:

$$A \cup B = \{0, 2, 4, 6, 8\} \cup \{0, 1, 2, 3, 4\} = \{0, 1, 2, 3, 4, 6, 8\}$$

$$(A \cup B) \cup C = \{0, 1, 2, 3, 4, 6, 8\} \cup \{0, 3, 6, 9\} = \{0, 1, 2, 3, 4, 6, 8, 9\}$$

$$A \cap B = \{0, 2, 4, 6, 8\} \cap \{0, 1, 2, 3, 4\} = \{0, 2, 4\}$$

$$(A \cap B) \cap C = \{0, 2, 4\} \cap \{0, 3, 6, 9\} = \{0\}$$

Computer Representation of Sets

Assume that the universal set U is finite.

First, specify an arbitrary ordering of the elements of U , for instance $a_1, a_2, \dots, a_n$.

Represent a subset A of U with the bit string of length n , where the i th bit in this string is 1 if a_i belongs to A and is 0 if a_i does not belong to A .

Example: Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, and the ordering of elements of U has the elements in increasing order. What bit strings represent the subset A of all odd integers in U , the subset B of all even integers in U , and the subset C of integers not exceeding 5 in U ?

Solution:

$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$A = \{1, 3, 5, 7, 9\}$$

$$B = \{2, 4, 6, 8, 10\}$$

$$C = \{1, 2, 3, 4, 5\}$$

Bit strings

$$U = 11\ 1111\ 1111$$

$$A = 10\ 1010\ 1010$$

$$B = 01\ 0101\ 0101$$

$$C = 11\ 1110\ 0000$$

Example: The universal set is $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. The given set is $A = \{1, 3, 5, 7, 9\}$. What is the bit string for $\bar{A}$?

Solution:

$$U = 11\ 1111\ 1111$$

$$A = 10\ 1010\ 1010$$

$$\bar{A} = 01\ 0101\ 0101 \text{ (Bit-wise complement)}$$

$$\bar{A} = \{2, 4, 6, 8, 10\}$$

Example:

The universal set is $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$.

Two sets are $A = \{1, 2, 3, 4, 5\}$ and $B = \{1, 3, 5, 7, 9\}$.

Use bit strings to find $A \cup B$, $A \cap B$, and $A - B$.

Solution:

$$U = 11\ 1111\ 1111$$

Union:

$$A = \quad 11\ 1110\ 0000$$

$$B = \quad 10\ 1010\ 1010$$

 $A \cup B = 11\ 1110\ 1010$ (bit-wise OR)

The corresponding set is $A \cup B = \{1, 2, 3, 4, 5, 7, 9\}$

Intersection:

$$A = \quad 11\ 1110\ 0000$$

$$B = \quad 10\ 1010\ 1010$$

 $A \cap B = 10\ 1010\ 0000$ (bit-wise AND)

The corresponding set is $A \cap B = \{1, 3, 5\}$

Example (contd):

The universal set is $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$.

Two sets are $A = \{1, 2, 3, 4, 5\}$ and $B = \{1, 3, 5, 7, 9\}$.

Use bit strings to find $A \cup B$, $A \cap B$, and $A - B$.

Solution:

$$U = 11\ 1111\ 1111$$

Difference:

$$A = 11\ 1110\ 0000$$

$$B = 10\ 1010\ 1010$$

$$\bar{B} = 01\ 0101\ 0101 \text{ (Bit-wise complement)}$$

$$A = 11\ 1110\ 0000$$

$$\bar{B} = 01\ 0101\ 0101$$

$$A - B = A \cap \bar{B} = 01\ 0100\ 0000 \text{ (Bit-wise AND)}$$

The corresponding set is $A - B = \{2, 4\}$